

Tutorial- 03

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Q1. Write four different C statements that each add 1 to integer variable x.

- `x++;`
- `++x;`
- `x += 1;`
- `x = x+1;`

Q2. Write a single C statement to accomplish each of the following:

a)Assign the sum of x and y to z and increment the value of x by 1 after the calculation.

- `z = x++ + y ;`

b)Multiply the variable product by 2 using the `*=` operator.

- `z *= 2 ;`

c)Multiply the variable product by 2 using the `=` and `*` operators.

- `z = z*2 ;`

d)Test if the value of the variable count is greater than 10. If it is, print “Count is greater than 10.”

- `If(count > 10) Printf (“Count greater than 10”);`

e)Decrement the variable x by 1, then subtract it from the variable total.

- `Total -= --x ;`

f)Add the variable x to the variable total, then decrement x by 1.

- `Total += --x ;`

g) Calculate the remainder after q is divided by divisor and assign the result to q. Write this statement two different ways.

- `q = q % divisor ;`
- `q %= divisor ;`

h) Print the value 123.4567 with 2 digits of precision. What value is printed?

- 123.45

i) Print the floating-point value 3.14159 with three digits to the right of the decimal point.

What value is printed?

- 3.142

Q3. Write single C statements that

a) Input integer variable x with scanf.

- `scanf("%d", &x) ;`

b) Input integer variable y with scanf.

- `scanf ( "%d" ,&y ) ;`

c) Initialize integer variable i to 1.

- `int i = 1 ;`

d) Initialize integer variable power to 1.

- `int power = 1 ;`

e) Multiply variable power by x and assign the result to power.

- `power *= x ;`

f) Increment variable i by 1.

- `i ++ ;`

g) Test i to see if it's less than or equal to y in the condition of a while statement.

- `while ( i <= y ) ;`

h) Output integer variable power with printf.

- `printf ( "%d " , power ) ;`